Andrew Collignon

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EDUCATION

University of Tennessee, Knoxville, Tickle College of Engineering, Knoxville, TN
B.S. in Mechanical Engineering, Expected May 2027
GPA: 3.35 / 4.00 | **Dean's List:** Fall 2023 – Spring 2024

RESEARCH EXPERIENCE Innovation South, University of Tennessee (METAL Program)Knoxville, TN

Undergraduate Research Assistant May – August 2026

- Designed and fabricated custom end effectors enabling Franka Research 3 and Universal Robots UR12e cobots to handle tools for forging and casting demonstrations.
- Designed a SpiRobs-inspired adaptive soft gripper for the UR12e: a 24-segment, tapered, cable-driven arm that wraps around irregular parts instead of clamping them.
- Collaborated with a three-person undergraduate robotics team on mechanical design, integration, and testing.

PROJECTS

4-Axis Robotic Arm Personal Project

- Designed and built a 4-axis arm from scratch in Fusion 360, with custom 20:1 cycloidal reducers on NEMA 17 steppers.
- Integrated per-joint ESP32-S3 controllers (TMC2209 drivers, AS5600 magnetic encoders) running micro-ROS firmware developed in PlatformIO.
- Achieved closed-loop position control of all four joints over ROS2 Jazzy, debugging DDS, USB-PD power, and serial issues.

BattleBots Featherweight Combat Robot – Lead Designer, Senior Design, 2026 – 2027

- Leading design of a 30 lb 4WD vertical spinner; previously designed a 3 lb beetleweight combat robot.

Clinostat for Microgravity Research – Lead Designer, Mississippi State University Collaboration

- Designed clinostat components for a study of microgravity effects on black soldier flies; coordinated fabrication and testing.

PROFESSIONAL EXPERIENCE

I.C. Thomasson Associates, Inc., a Salas O'Brien Company, Nashville, TN
Mechanical Engineering Intern May – August 2025

- Designed HVAC ductwork and piping for healthcare and residential buildings; produced design sheets and construction documents.
- Conducted on-site field visits to verify construction progress and gather system data for design updates.

Envision Advantage, LLC, Nashville, TN
Mechanical Engineering Intern May – August 2024

- Designed ductwork and piping for hospitals and schools in Revit, working with engineers to meet code and project specifications.
- Performed airflow load calculations and mechanical air-balance analysis to support HVAC system design.

LEADERSHIP & ACTIVITIES

Ynot Robotics Club, University of Tennessee Jan 2023 – May 2025
Vice President (Oct 2023 – May 2025); Design Lead (Jan 2023 – May 2025)

- Led a team of 7 through mechanical design reviews and iterations; machined parts on lathes and mills.
- Secured sponsorships and funding, and built budgeting spreadsheets to allocate funds and track project expenses.

High School Competitive Robotics

- 3× consecutive state champion, Canadian Open Signature Event winner, 4× Worlds qualifier, and Worlds Design and Build award winner.

SKILLS

- **CAD & Design:** Fusion 360, SolidWorks, Autodesk Revit
- **Robotics & Programming:** ROS2, micro-ROS, Python, MATLAB, embedded C++ (ESP32, Arduino), Linux/WSL2
- **Fabrication:** FDM and SLA 3D printing, lathe, mill, electronics integration
- **Robot Platforms:** Universal Robots UR12e, Franka Research 3